PROPOSITIONAL LOGIC

CSE 2213 – Discrete Mathematics

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WHAT IS A PROPOSITION?

- A proposition is a declarative sentence
  - It can be either true or false
  - It cannot be neither
  - It cannot be both

Example: Today is Sunday

It's raining outside.

WHAT IS NOT A PROPOSITION?

- What is your name?
  - A question. Not a declarative sentence.
- Tell me your name.
  - An order. Not a declarative sentence.
- $x + 1 = y$
  - An equation. Neither true nor false.
- $x + 1 = y$, given that $x = 5$ and $y = 6$
  - An equation with values of each variable. This is a proposition.

PROPOSITION

- We sometimes use symbols to represent a proposition, without writing the actual sentence.
- $p =$ My name is Mahim.
  - Here, p represents "My name is Mahim"
- $q =$ Today is Sunday.
  - Here, q represents ""Today is Sunday"

COMPOUND PROPOSITIONS

- My name is Mahim and I am a lecturer.
- We can express this proposition as a combination of two propositions, along with a linker
- Such linkers are – and, or, if, only if, if and only if

LOGICAL CONNECTIVES

- OR (∨)
- AND (∧)
- Negation/ NOT (¬)
- Implication / if-then (→)
- Biconditional (⇔)

COMPOUND PROPOSITIONS

- Consider the following proposition: My name is Mahim.
- My name is not Mahim
- Today is Sunday
- Today is not Sunday
- It's not the case that today is Sunday.

It only negates the actual proposition

- Such a proposition is called a negation

NEGATION

- p: Today is Monday
- $\neg p$:

Today is not Monday.

It's not the case that today is Monday.

- True if the actual proposition is false.

p	$\neg p$
T	F
F	T

CONJUNCTION

- Today is Monday and it's raining outside.
- $p \wedge q$
- True if both propositions are true

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

DISJUNCTION

- Today is Monday or it's raining outside.
- $p \lor q$
- True if any proposition is true

p	q	$p \lor q$
T	T	T
T	F	T
F	T	T
F	F	F

IMPLICATION/CONDITIONAL STATEMENT

- If it rains, then I will stay home.
- Let P and Q be propositions.
- P: It rains
- Q: I will stay home
- P is called premise
- Q is called conclusion/consequence

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

BICONDITIONAL

- P: You can take the flight
- Q: You buy a ticket
- You can take the flight if and only if you buy a ticket.

P	Q	$P \Leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

A SPECIAL TYPE OF DISJUNCTION – EXCLUSIVE-OR

- I will give you laptop or mobile.
  - Cannot have both!!!
- $p \oplus q$
- True if exactly one proposition is true

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

TRUTH TABLES OF COMPOUND PROPOSITIONS

- The propositions of a compound statement may have different truth values
- In a truth table, we consider all truth values of the propositions, and find out the truth value of the compound one

p	q	r	s
T	T	T	?
T	T	F	?
T	F	T	?
T	F	F	?
F	T	T	?
F	T	F	?
F	F	T	?
F	F	F	?

LOGICAL OPERATOR PRECEDENCE

Operator	Precedence
$\neg$	1
$\wedge$ $\vee$	2 3
$\rightarrow$ $\leftrightarrow$	4 5

TRUTH TABLE

- Construct truth table for the following statements
- $Q \vee (\neg Q \wedge \neg P)$
- $(A \vee B) \wedge \neg (B \vee C)$
- $(P \vee \neg Q) \wedge (\neg P \vee Q)$

$$(A \rightarrow B) \wedge \neg (A \rightarrow B)$$

A	B	A->B	¬(A->B)	(A->B) ∧¬(A->B)
T	T	T	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	F	F

EXERCISE

Construct a truth for each of the following compound propositions.

- $(A \rightarrow B) \Leftrightarrow \neg(A \oplus B)$
- $P \rightarrow Q \land (\neg Q \land \neg P)$
- $P \land (Q \Leftrightarrow R)$

P ∧ (Q ⇔ R)

P	Q	R	Q<->R	P Λ (Q<->R)
T	T	T	T	T
T	T	F	F	F
T	F	T	F	F
T	F	F	T	T
F	T	T	T	F
F	T	T	T	F
F	F	T	F	F
F	F	F	T	F

DIFFERENT WAYS OF EXPRESSING CONDITIONAL STATEMENT

if p, then q
if p, q
q unless $\neg p$
q if p
p is sufficient for q
q is necessary for p
a sufficient condition for q is p

p implies q
p only if q
q when p
q whenever p
q follows from p
a necessary condition for p is q

BICONDITIONAL STATEMENT

- **Different ways to express p $\Leftrightarrow$q :**
- p if and only if q
- p iff q
- p is necessary and sufficient for q
- if p then q, and if q,then p

INVERSE, CONVERSE, CONTRAPOSITIVE

- $q \rightarrow p$ is the **converse** of $p \rightarrow q$
- $\neg q \rightarrow \neg p$ is the **contrapositive** of $p \rightarrow q$
- $\neg p \rightarrow \neg q$ is the **inverse** of $p \rightarrow q$
- Find the converse, inverse, and contrapositive of "The football team wins whenever it is raining.“
- Rewriting the sentence : "If it is raining, then the football team wins."

➢ *$q \rightarrow p$: If the football team wins, then it's raining.*

➢ *$\neg q \rightarrow \neg p$: If the football team doesn't win, then it's not raining*

➢ *$\neg p \rightarrow \neg q$: If it's not raining, then the football team doesn't win.*

TRANSLATING ENGLISH SENTENCES

- How can this English sentence be translated into a logical expression?
  " If you submit the report, then you will get bonus mark.
- Let p and q be a proposition:
- p: You submit the report
- q: You get bonus mark
- Logical expression: $p \rightarrow q$

TRANSLATING ENGLISH SENTENCES

- Chapter 1: 1.1
- Exercise: 7, 9, 10

TRANSLATING LOGICAL EXPRESSION TO ENGLISH

- Chapter 1: 1.1
- Exercise : 4, 5, 6, 8

FIND THE TRUTH TABLE : $P \to Q$, $Q \to P$,$\neg Q \to \neg P$, $\neg P \to \neg Q$

P	Q	¬P	¬Q	P->Q	Q->P	¬Q->¬P	¬p->¬q
T	T	F	F	T	T	T	T
T	F	F	T	F	T	F	T
F	T	T	F	T	F	T	F
F	F	T	T	T	T	T	T

PRACTICE PROBLEM

- Find the inverse, converse and contrapositive of the following sentence.

  I come to class whenever there is going to be a quiz.